Mahesh Moholkar

Full-Stack Developer

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Summary

Full-Stack Developer (3+ years) on .NET and React: built a Campus ERP for a university and its affiliated colleges serving 8000+ users, with shared auth and notification microservices – plus LLM projects on Python/FastAPI and Next.js (RAG helpdesk assistant, service desk with LLM ticket routing, natural-language reporting, document extraction) with evals in CI.

Professional Experience

Full-Stack Developer **Aug 2023 – Present** · *3 yrs 3 mos*

Shivam Enterprises *Pune, India*

- Built a **Campus ERP for a university and its affiliated colleges** on **.NET 10 + React** – one platform with colleges as tenants, serving **8000+ users**; brought p95 under 400 ms at k6 peak load with Redis HybridCache.
- Built two separate LLM projects on **Python/FastAPI** (PostgreSQL + pgvector): a **helpdesk assistant** answering from policies and circulars with citations, and **Campus Support**, a Next.js service desk for staff requests to the dev team where an LLM checks ticket completeness, classifies requests and suggests the assignee with confidence gating; added natural-language report queries and document extraction to the main ERP.
- Built the **RAG and evaluation pipeline**: hybrid retrieval (pgvector + BM25, reciprocal-rank fusion, reranking), tool calls with per-user access checks for personal data, golden-set evals with LLM-as-judge in CI.
- Built shared **auth** (JWT, role- and menu-based access control) and **notification** (outbox + Hangfire workers: SendGrid email, SMS and push with retries, delivery tracking) **microservices** used by every ERP.
- Built separate **Admissions, Management and Procurement ERPs** (7-stage admissions pipeline; directors' console across all colleges; vendor portal, quotations, indents), each with its own database – React UI, .NET 10 REST APIs, EF Core schema and migrations – using the shared auth and notification microservices.
- Built the **React/TypeScript UI** for the main ERP's exams, fees, HR, attendance, LMS and research modules with typed API clients; worked with the backend team on API contracts.
- Set up Serilog + correlation IDs, OpenTelemetry/Grafana, a HealthChecks dashboard and **Azure DevOps CI/CD** with Testcontainers tests and Docker builds; wrote ADRs and the onboarding runbook; mentored interns.

Stack: .NET 10, React, TypeScript, SQL Server, Redis, Python, FastAPI, Next.js, PostgreSQL, pgvector, LLM APIs, Docker, Azure DevOps

Frontend Developer (Part-time, Remote) **Jan 2025 – Jun 2026** · *1 yr 6 mos*

Vision – Multi-tenant SaaS ERP for GCC SMEs *Kuwait*

- Built the **React 18 + TypeScript client** (Vite) for a multi-tenant GCC ERP – quotation, sales order, invoice, returns, purchase order and bill screens, inventory and VAT-aware totals – as typed forms (React Hook Form + Zod) with state-driven edit/post/void actions.
- Owned the client data layer: TanStack Router (loaders, search-param filters), TanStack Query with per-document cache invalidation and optimistic updates, Zustand for session state, real-time notifications over WebSockets.
- Shipped a bilingual **English/Arabic RTL-first UI** (i18next, shadcn/Radix + Tailwind design system), permission-manifest-driven navigation, printable document views, and a shared component library reused across the web app, public storefront and admin app.
- Tested with **Vitest + Testing Library** and Playwright; GitHub Actions CI with Docker builds deployed on **AWS**.

Stack: React 18, TypeScript, Vite, TanStack Router, TanStack Query, Zustand, React Hook Form, Zod, Tailwind CSS, shadcn/Radix, i18next

Fellowship – Zero to Merge Program **Aug 2023 – Sep 2023** · *2 mos*

Cloud Native Computing Foundation (CNCF) *Remote*

- CNCF mentored open-source program (Incubator Badge): PRs to Kubernetes-ecosystem projects (Go); bug fix merged into **Ockam** (Rust); documentation contributions to Kubernetes SIGs and EvalML.

Skills

Backend: Python (FastAPI, Pydantic), C#/.NET 10 (EF Core, SignalR, Hangfire), REST APIs (OpenAPI), JWT auth, RBAC, SendGrid, Go

Frontend: React, Next.js, TypeScript, TanStack Query, TanStack Router, Zustand, React Hook Form, Zod, Tailwind CSS, i18next, Expo

AI/ML: LLM APIs (tool use, structured outputs, prompt caching), RAG (embeddings, hybrid retrieval, reranking), NL-to-SQL, evals

Data & Infra: SQL Server, PostgreSQL, pgvector, Redis, Docker, AWS, Azure DevOps, GitHub Actions, Railway, Serilog, OpenTelemetry, Grafana, xUnit, pytest, Testcontainers, Vitest, k6, Playwright

Projects

Vyavasay AI ERP – GST Billing & Inventory SaaS for Indian SMEs (2026) **vyavasayapp.in**

- Multi-tenant ERP: quote-to-cash and purchase cycles, GST (CGST/SGST/IGST, HSN, GSTR returns), double-entry GL, multi-location inventory, storefront and Expo mobile app; **LLM bill OCR** and **pgvector** image search; Testcontainers integration tests.

Stack: .NET 10, EF Core, React, TypeScript, TanStack Router, Expo, PostgreSQL, pgvector, Hangfire, SignalR, LLM APIs, Docker, Railway

Education

Dr. D. Y. Patil Institute of Management and Research **2022 – 2024**

Master's in Computer Application – CGPA: 8.0 / 10 *Pune, India*